

CS536 - Science of Programming

Fall 2025 Syllabus

for Sections 01, 02, 03 and 04

Xiaolang Wang

August 2025

1 Course Summary

Welcome to CS-536 - Science of Programming. This is a theoretical computer science graduate course about program verification: we will use a simple programming language to show how to design and prove reliable programs. The class starts on August 18, and lectures will be given on Wednesdays and Fridays 10:00 - 11:15 am, at SB 104. Students in Section 03 (online section) are highly encouraged to attend lectures in person. Students in Section 04 (Beacon section) will have access to lecture recordings on Canvas and on Lumina.

2 Prerequisites and Course Outcome

- The prerequisites for this class are propositional logic and predicate logic (covered in CS-330 and maybe CS-401).
- At the end of the class, the students should be able to answer the following questions:
 1. How does verification differ from testing?
 2. How do we understand the semantics of programs?
 3. How do semantics affect the way we understand the correctness of the program?
 4. What rules of reasoning can we use to discuss the correctness of the program?
 5. How is the correctness of parallel programs different?
 6. What if program execution is not deterministic?

3 Textbook

There is no textbook required for this course. Class notes will be provided before each class. I will also share the handouts by Professor James Sasaki on the Canvas as a reference.

A helpful book is *The Science of Programming* by David Gries (ISBN-10: 0387964800 — ISBN-13: 978-0387964805).

Any Discrete Math textbook that covers propositional logic and predicate logic can also be helpful.

4 Getting Help

- Instructor: Xiaolang Wang's office hours are on Thursdays 12:00 - 3:00 pm at SB 208-D; no appointment needed, first-come-first-serve. If extra help is needed, please email xwang122@illinoistech.edu, I will respond to emails every weekday.
- TAs:
 - Chaoqi Ma (cma17@hawk.illinoistech.edu), office hours: Mondays 1:00 - 2:00 pm at SB 004.
 - Xu Yang (xyang99@hawk.illinoistech.edu), office hours: Tuesdays 10:00 - 11:00 am on Zoom at
<https://iit-edu.zoom.us/j/81216915964?pwd=49ECsAMCFrYxOqLpuJXGcZozr1e1P9.1>

We will use Zoom on the Canvas for this class so that students can easily access the lectures online. Students can access the online lectures on the Canvas: **CS536 course page** → **Zoom** → **Upcoming Meetings**.

Students can also find the recordings of each class on the Canvas: **CS536 course page** → **Panopto Videos**. These recordings will also be uploaded to Lumina for students in Section 04.

Class notes and partial homework solutions will be posted on Canvas (and also on Lumina for students in Section 04).

Students are expected to check their email every weekday of the semester. Clarifications on assignments or other important announcements might be sent by email - and will also be posted on the Canvas.

5 Grading and Assignments

The grading allocation is given below.

- Homework Assignments $6\% \times 6$
- Midterms $18\% \times 2$
- Final 28%

Assignments

There will be six homework assignments (most likely). In each assignment, students will see questions similar to the examples in class. Each student is expected to write their own solution, but discussion between students is highly encouraged. Do not use any online resources (including ChatGPT) other than the classnotes for homework. A student can seek help from me and TA's if they are having difficulties with the homework. But, spend some time trying to understand a homework problem before asking for help.

Each assignment is expected to be submitted on Canvas before its deadline. (Students in Section 04 submit assignments to Lumina instead.)

The penalty for late submission is 20% for each day after the deadline. If you have difficulty finishing an assignment on time, email me **before the deadline** and we can discuss extension. Each assignment has a 20% “early bird” extra credit if it is submitted at least two days before the deadline.

Exams

The midterms and the final for students in Section 01, 02 and 03 will be **in-person**, no electronic devices (including calculators) and closed-books-closed-note with the exception of a “cheat sheet”. An A4-sized double-sided cheat sheet is allowed for each midterm and three are allowed for the final. Students are expecting multiple choice questions and short answer questions similar to class examples and questions in homework assignments in each exam. **Midterm 1 will be in class on October 1, Midterm 2 will be in class on November 12** and the final will take place in the final exam week.

Important: In each exam, there will be one (or more) additional questions for Section 02 (PHD) students. Other students can also answer these questions for extra credits.

Important: Students in Section 03 (online section) should also take exams in person. If a student is located in Chicago, the student is encouraged to attend the exam together with other students in class. If a student is located in other cities in the US, we will work together with the Center for Learning Innovation to find an exam center close to the student; in the case, please email me and Charles Scott (scott@illinoistech.edu) from the Center **at least 2 weeks** before each exam to schedule this service. If a student is in Chicago but cannot attend the exam at the scheduled time, email me **at least 2 weeks** before each exam to find an alternative time for that exam.

Important: For students in Section 04, exams will take place through Lumina. **Midterm 1 will be on Friday October 3 Midterm 2 will be on Friday November 14** and the final will take place in the final exam week. More details about the exams will be posted in the future on Lumina.

The final grades will be assigned by comparison to the students who took this class with me in previous and current semesters. As a guideline, about 80% will be needed for an A, and 65% for a B; these lines might change. So far, 224 students have got an A, 273 students have got a B, 299 students have got a C, and 17 students have got an E.

The standard departmental policy regarding academic (dis)honesty applies. See the IIT Code of Academic Honesty, <https://web.iit.edu/student-affairs/handbook/fine-print/code-academic-honesty>. Engaging in any form of academic dishonesty during the exam will result in an assigned letter grade of E for the student involved.

6 Lecture Schedule

Week	Topics	Notes
#1 (Aug 20 and 22)	Syllabus, Propositional Logic, Predicate Logic	HW1 assigned
#2 (Aug 27 and 29)	Expressions, States, Satisfaction and State Updates	
#3 (Sep 3 and 5)	Language Syntax, Operational Semantics	HW2 assigned
#4 (Sep 10 and 12)	Denotational Semantics, Divergence, Runtime Errs	
#5 (Sep 17 and 19)	Hoare Triples	HW3 assigned
#6 (Sep 24 and 26)	Weakest Preconditions	
#7 (Oct 1 and 3)	Syntactic Substitution	Midterm 1 on October 1, topics: HW 1 to 3; HW4 assignment
#8 (Oct 8 and 10)	Strongest Postconditions and Forward Assignment	
#9 (Oct 15 and 17)	Proof Rules, Proof Outlines	HW5 assigned
#10 (Oct 22 and 24)	Partial and Minimal Proof Outlines, Total Correctness Rules	
#11 (Oct 29 and 31)	Loop Invariants	HW6 assigned
#12 (Nov 5 and 7)	Array Assignments, Basics of Parallel Programs	
#13 (Nov 12 and 14)	Disjoint Parallel Programs	Midterm 2 on November 12, topics: HW 4 to 6
#14 (Nov 19 and 21)	Disjoint Conditions	
#15 (Nov 26 and 28)	Shared Variables and Interference	No lecture on 28 for Thanksgiving
#16 (Dec 3 and 5)	Await and Deadlocks	
#17 (Dec 8 - 12)		Final date TBD, topics: everything discussed in this course

7 Administrative Matters

Reasonable accommodations will be made for students with documented disabilities. In order to receive accommodations, students must obtain a letter of accommodation from the Center for Disability Resources (CDR) located at 3424 S. State Street - 1C3-2, 312 567.5744 or disabilities@illinoistech.edu.

Illinois Tech's Sexual Harassment and Discrimination Information:

Illinois Tech prohibits all sexual harassment, sexual misconduct, and gender discrimination by any member of our community. This includes harassment among students, staff, or faculty. Sexual harassment of a student by a faculty member or sexual harassment of an employee by a supervisor is particularly serious. Such conduct may easily create an intimidating, hostile, or offensive environment.

Illinois Tech encourages anyone experiencing sexual harassment or sexual misconduct to speak with the Office of Title IX Compliance for information on support options and the resolution process.

You can report sexual harassment electronically at iit.edu/incidentreport, which may be completed anonymously. You may additionally report by contacting the Title IX Coordinator, Virginia Foster at foster@illinoistech.edu or the Deputy Title IX Coordinator at eespeland@illinoistech.edu.

For confidential support, you may reach Illinois Tech's Confidential Advisor at (773) 907-1062. You can also contact a licensed practitioner in Illinois Tech's Student Health and Wellness Center at student.health@illinoistech.edu or (312)567-7550

For a comprehensive list of resources regarding counseling services, medical assistance, legal assistance and visa and immigration services, you can visit the Office of Title IX Compliance website at <https://www.iit.edu/title-ix/resources>.